

4x3.5

7. Write short notes on any four -

- (1) Joule - Kelvin expansion
- (2) Corollaries of second law of thermodynamics.
- (3) Second law efficiency
- (4) Classification of air compressors.
- (5) First and second Tds equations
- (6) Non steady/Variable flow processes.

$$du = ds + dv$$

$$ds = \frac{du + dv}{T} \quad \text{--- (1)}$$

$$dh = du + v + dv$$

$$dh = du + v + dv + vdp$$

$$Tds + vdp + vdv + vdv$$

$$dh = Tds + vdp \quad \text{--- (2)}$$

$\frac{A_m v}{A}$ for perfect gas

$$Tds = Cv dt + T \left(\frac{\partial p}{\partial T} \right)_v dv$$

$$Tds = Cv dt - T \left(\frac{\partial v}{\partial T} \right)_p dp$$

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$$

$$\frac{T}{V} \frac{dp}{ds}$$

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✓ ✓ ✓

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Roll No.

BE II (ME)

3

ET

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME41A (ME) : ENGINEERING

THERMODYNAMICS (M)

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any Five questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) State Zeroth law of thermodynamics and Kelvin Planck statement. What is a Carnot cycle. Write the four

processes, which constitute the cycle. 7

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1

(Contd.)

(b) State Clausius statement of second law of thermodynamics

A cyclic heat engine operates between a source temperature of 800°C and a sink temperature of 30°C . Find the least rate of heat rejection per kW net output of the engine. 7

2. (a) Prove that maximum work obtainable from two finite bodies at temperatures T_1 and T_2 is $W_{\max} = C_p(\sqrt{T_1} - \sqrt{T_2})$ where symbols have usual meanings. 7

(b) What is energy? Calculate the decrease in energy when 25kg of water at 95° mix with 35kg of water at 35°C , the pressure being taken as constant and the temperature of the surrounding being 15°C (c_p of water = 4.2kJ/kg K) 7

3. (a) State Dalton's law of partial pressure, Amagat's law of partial volumes and Gibbs's theorem. 7

(b) Write ideal gas equation. A fluid at 200kPa and 300°C has a volume of 0.8m^3 . In a frictionless process at constant volume the pressure changes to 100kPa. Find the final temperature and the heat transferred if the fluid is air. Take R for air as 0.287kJ/kg K . 7

4. (a) Derive four Maxwell's equations. 3305 7

(b) Derive the following Clausius - Clapeyron equation -

$$\frac{dP}{dT} = \frac{h_{fg}}{T(v_g - v_f)} \text{ write the use of the equation. } 7$$

5. (a) What is stoichiometric air fuel ratio and excess air? Write combustion equation for (1) C (Carbon), (2) H_2 , (3) CH_4 (Methane) & (4) S (sulfur) fuels where fuels are burnt with chemically correct amount of air at atmospheric pressure. 7

(b) A fuel having formula C_7H_{16} (Heptane) is burnt with chemically correct amount of air at atmospheric pressure. Write the combustion equation and find the stoichiometric air fuel ratio. 7

6. (a) Define and derive the expression of volumetric efficiency of a single stage compressor with clearance. Also discuss how the volumetric efficiency varies with clearance and pressure ratio. 7

(b) Derive the expression of optimum intermediate pressure of a two stage reciprocating compressor for minimum work considering perfect intercooling and perfect after-cooling condition. 7

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Roll No. [REDACTED]

BE II (ME)

3

MT

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME42A (ME) : MATERIALS TECHNOLOGY

Time - Three Hours

Maximum Marks - 70

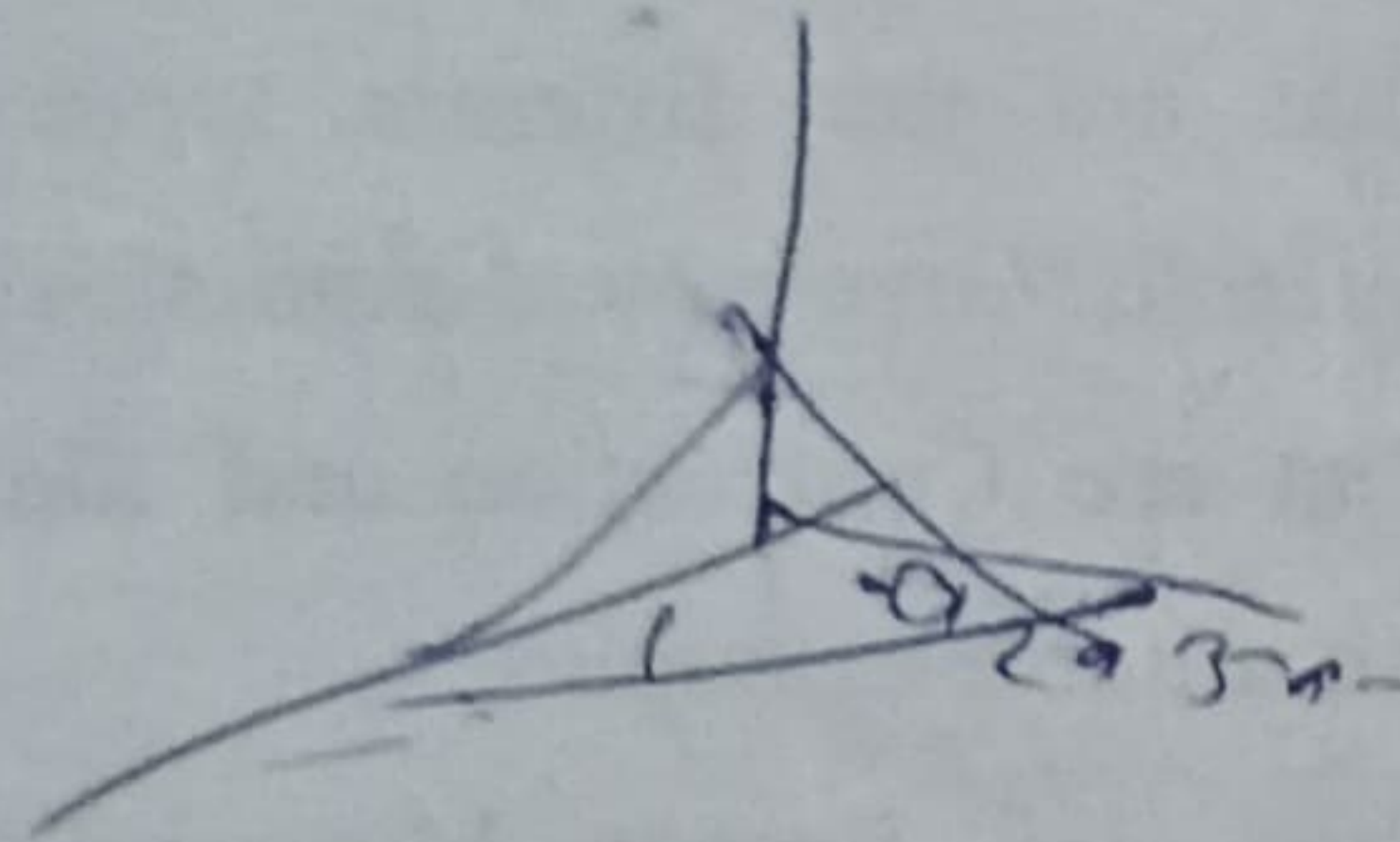
- Note:- (i) Attempt any **Five** questions.
(ii) Answer should be to the point.
(iii) All questions carry equal marks.

1. (a) What type of lattice structure is usually indicative of ductility in metals? 3
- (b) A cube has six equivalent faces. Assuming a definite orientation of the cube, write down the Miller indices of the six facial planes of the cube. 3
- (c) A metal has a BCC crystal structure and an atomic radius of 1.732 Å. Calculate the number of atoms per square centimeter of (100) planes. 4

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1

(Contd.)



3 2 29

2. Explain the difference between the following pair of terms:-
- (a) Elastic limit and proportional limit 3
 - (b) Toughness and resilience 4
 - (c) Elasticity and plasticity 3
3. (a) Define: Ferrite, Cementite, Austenite, Pearlite, Martensite and Bainite. What are their general characteristics? How are they obtained? 10 749 7
- (b) What is meant from 'Critical Temperature Range'? 3
- What practical significance does it have? 3
4. (a) What is the major difference in the purpose of annealing and normalizing? 4
- (b) What is the importance of hardenability of steel as applied to heat treatment? 3
- (c) How do the composition and structure of a high strength constructional steel differ from that of a tool steel? 3
5. (a) High-carbon-high-chromium steels are widely used in industry. What are their important characteristics and applications? 4
- (b) How would you control the composition of cast iron produced by Cupola? 3

- (c) What are the advantages of aluminium alloys? Where are they used? 3
6. (a) Comment upon the metallurgical structure of bearing alloys and compare it with that of metals and homogeneous alloys. 3
- (b) What are Refractory Materials? Give a few examples of refractory materials. 4
- (c) What are the different types of composites available? Give typical examples of each group. 3
7. (a) What are Compaction and Sintering? Explain with suitable examples. 4
- (b) What are Smart Materials? Give their applications. 3
- (c) Differentiate between Nitriding and Cyaniding. 3

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Roll No.

BE II (ME)

3

KOM

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME43A (ME) : KINEMATICS OF MACHINES

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Differentiate between Mechanism and Machine. 4

(b) What do you meant by Inversion of mechanism?

Explain any two inversions of double slider crank chain mechanism. 10

J-1435

1

(Contd.)

2. A quick return mechanism is shown in fig. 1. Link 2 rotates uniformly at 20 rad/sec. Draw the velocity diagram and acceleration diagram and obtain the total acceleration of slider B.

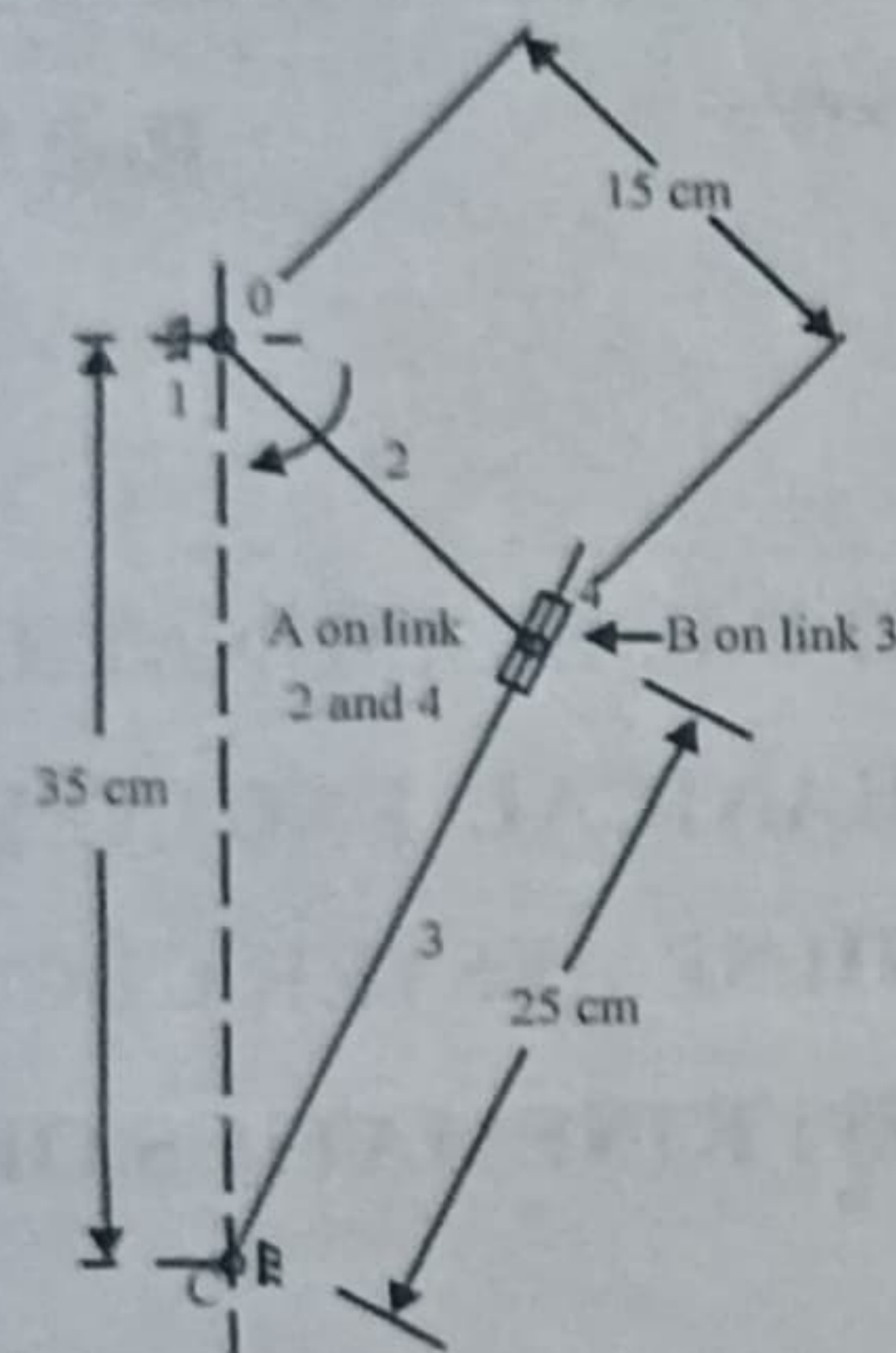


Fig. 1

3. (a) Explain the Pantograph mechanism with neat sketch. 7
 (b) For the Universal joint obtain the expression for ratio of speed of driving and driver shaft. 7
4. (a) What is the limiting angle of friction? Explain. 4
 (b) Calculate power lost in overcoming friction and number of collars required for the thrust bearing whose contract surfaces are 20 cm external radius and 15 cm. in internal radius. The coefficient of friction is 0.08. The total axial load is 30 kN. Intensity of pressure is not to exceed 3.5 bar. Speed of shaft is 420 rpm. 10

5. (a) What are the advantages of rolling contact bearings over other bearings? 4
 (b) A car engine has its rated output of 10kw. Maximum torque developed is 100 N-m. The clutch used is of single plate type, having two active surfaces. Axial pressure is not to exceed 0.85 bar. External diameter of the friction plate is 1.25 times the internal diameter. Determine the dimensions of the friction plate and the axial force exerted by the springs. Assume $\mu = 0.3$. 10
6. (a) What is creep in belts? What is its effect in belt drive? 4
 (b) A leather belting mass 1 gm/cm^3 has a maximum permissible tension of 21 bar. Determine the maximum power that can be transmitted by a belt $25\text{cm} \times 1.1 \text{ cm}$, if the ratio of friction tension is 2.14302 10
7. (a) Sketch a Bush and Roller Chain. Give the classification of chains. 4
 (b) 100 kw is transmitted by rope drive through a 160 cm diameter 45° grooved pulley running at 200 rpm. Angle of overlap is 140° and coefficient of friction between the pulley and the rope is 0.25. Mass of rope is 0.7 kg/m. and it can withstand a tension of 800 N. Considering the centrifugal tension induced, determine
 (i) the number of ropes needed for transmitting the required power.
 (ii) the tension in the rope before starting. 10

6. (a) A steel bar of rectangular section $30 \text{ mm} \times 40 \text{ mm}$ one end fixed and other is free at each end is subjected to axial compression. The bar is 1.75 m long. Determine the buckling load and the corresponding axial stress using Euler's formula.

Determine the minimum length for which Euler's equation may be used to determine the buckling load, if the proportional limit of the material is 200 N/mm^2 .

Take $E = 2 \times 10^5 \text{ N/mm}^2$. 14

7. Two planes AB and BC which are at right angles carry shear stresses of intensity 35 N/mm^2 while these planes also carry a tensile stress of 140 N/mm^2 and a compressive stress of 70 N/mm^2 respectively. Determine the principal planes and the principal stresses. Also determine the maximum shear stress and the planes on which it acts. 14

Ans. 154.492
1m 55.50

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Roll No.

BE II (ME)

3

Mech. of Solids

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME44A : MECHANICS OF SOLIDS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Two copper rods and one steel rod together support a load 500 kN as shown in Fig. 1. Find the stresses in the rods. Take $E_s = 2 \times 10^5 \text{ N/mm}^2$ and $E_c = 1 \times 10^5 \text{ N/mm}^2$.

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1

(Contd.)

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4

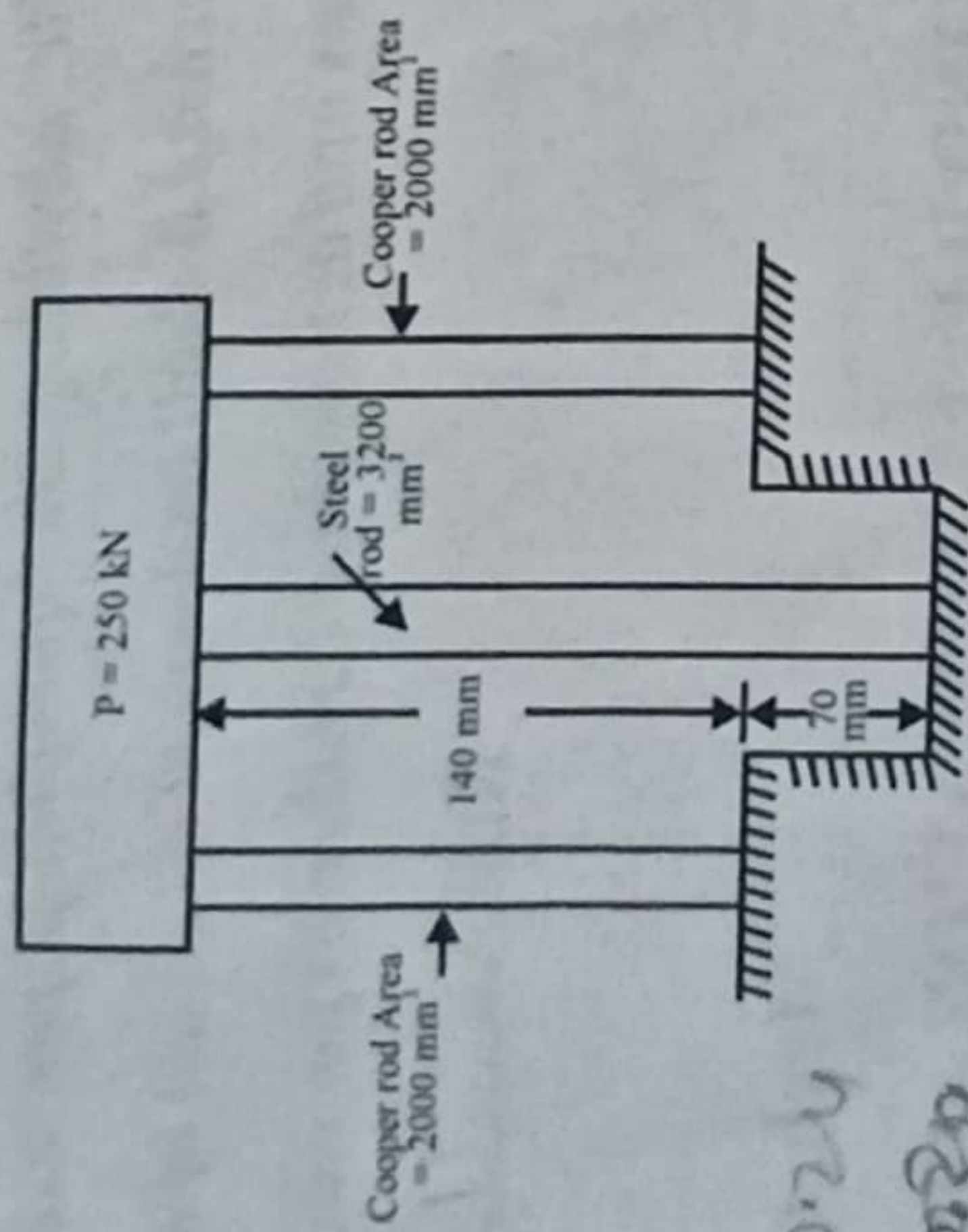


Fig. 1

10

(b) A steel bar of 30 mm diameter is loaded as shown in

Fig. 2. Determine the stresses in each part and the total elongation. Take $E = 210 \text{ GPa}$.

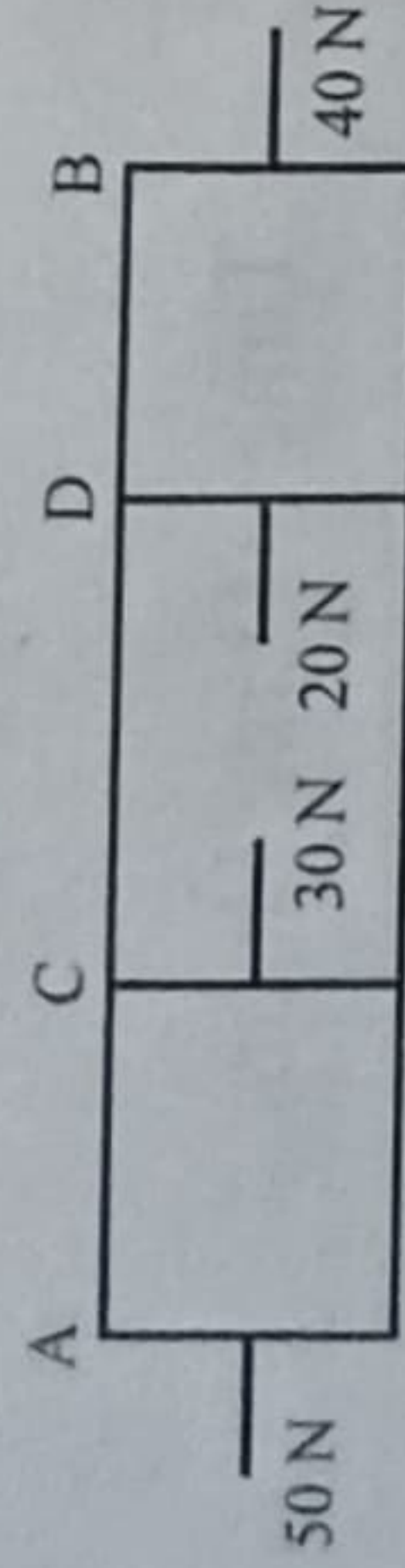


Fig. 2

4

2. Draw SFD and BMD for the beam shown in Fig. 3

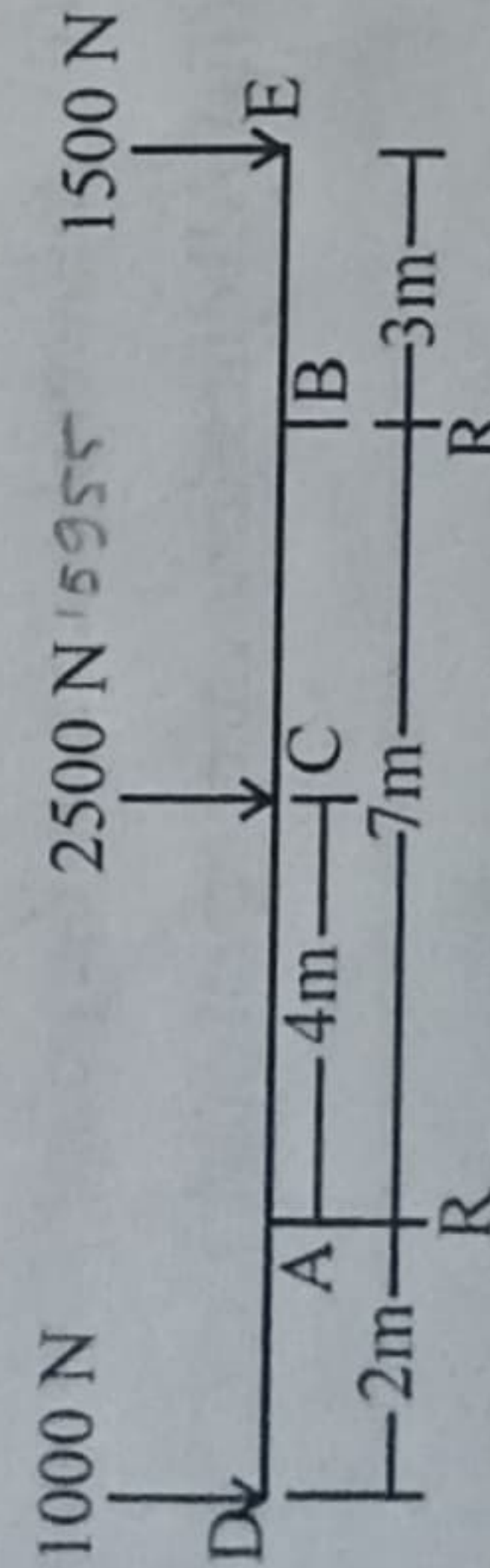


Fig. 3

14

3. (a) A timber joist of 200 mm width and 400 mm depth is used as a cantilever beam of 3 meters length and loaded

with a concentrated load at its free end in addition to its self-weight, so that the bending stress does not exceed 7 N/mm^2 . Find out the maximum value of the applied load. Unit weight of timber is 5 kN/m^3 .

10

(b) What do you mean by Section Modulus? Find an expression for section modulus for rectangular, circular and hollow circular section.

4

4. For the loaded beam shown in Fig. 4 find (i) the slopes at A, C and D (ii) the deflection at C and D and (iii) the maximum deflection. Take $E = 200 \text{ kN/mm}^2$ and $I = 9.25 \times 10^7 \text{ mm}^4$.

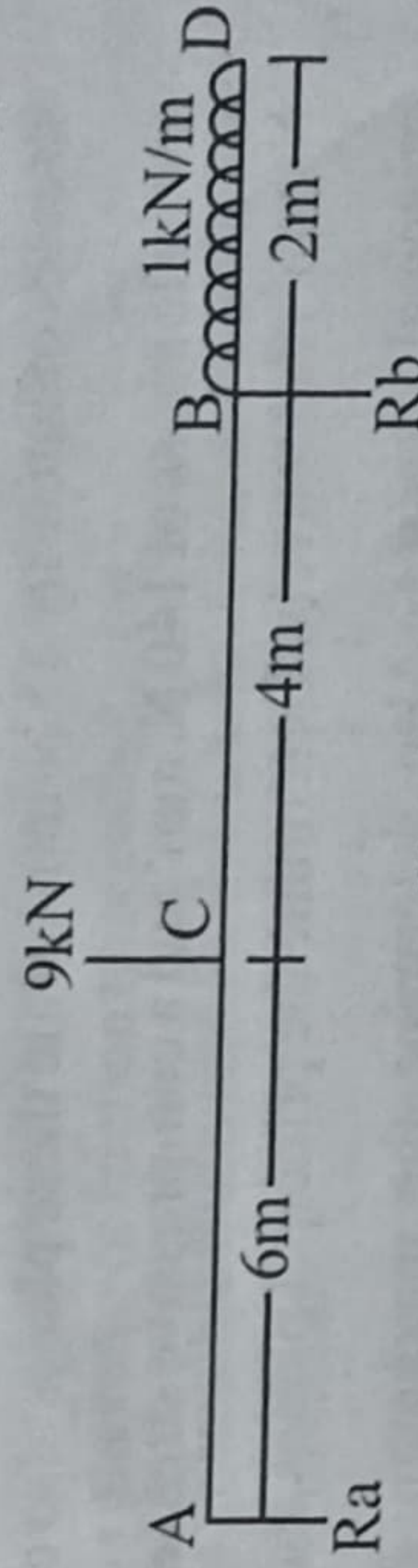


Fig. 4

14

5. (a) Explain significance of Rankine formula for Columns.

(b) A hollow shaft is to have an outside diameter d and inside diameter $\frac{d}{2}$. Calculate the minimum value of d if it is to transmit 375 kW at 100 rpm with a working stress of 40 N/mm^2 . Determine the twist in a length equal to 10 times the external diameter. Take $C = 8 \times 10^4 \text{ N/mm}^2$.

10

(b) Solve

$$(2x^2y + 2xy^2 + 2xyz + 1)dx + (x^3 + x^2y + x^2z + 2xyz + 2y^2z + 2yz^2 + 1)dy + (xy^2 + y^3 + y^2z + 1)dz = 0$$

7

2. (a) Solve the following Partial differential equation by using Charpit's Method.

$$px + qy = pq$$

7

(b) Solve the following Partial differential equation by using Monge's Method.

$$q^2r - 2pqs + p^2t = 0$$

7

3 (a) The ordinate of the normal curve are given by the following table

x :	0.0	0.2	0.4	0.6	0.8
y :	0.3989	0.3910	0.3683	0.3332	0.2897

Evaluate (i) $y(0.25)$ (ii) $y(0.43)$

7

(b) Using Lagrange's Interpolation formula, find the value of y when $x = 3$ from the following data :

x :	0	1	2	4	5
y :	0	16	48	88	0

7

4. (a) Compute the values of $\int_0^6 \frac{dx}{1+x^2}$ by the (i) Trapezoidal Rule. (ii) Simpson's $\frac{1}{3}$ Rule and Compare your result with the true value.

7

J-1437

2

(b) Find the real root of $2x - 3\sin x - 5 = 0$ upto six decimal places by Newton's Raphson Method.

7

5. (a) Solve the following system by Gauss - Seidel Iteration Method

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

7

(b) Using Runge - Kutta Method, find the approximate value of $y(0.2)$ if $\frac{dy}{dx} = x + y^2$ given that $y = 1$ when $x = 0$.

7

6. (a) Find Laplace $\sin \sqrt{t}$ also

$$\text{Find Laplace } \frac{\cos \sqrt{t}}{\sqrt{t}}$$

7

(b) State Convolution Theorem and apply the Convolution Theorem to obtain

$$\mathcal{L}^{-1} \left[\frac{s^2}{(s^2 + a^2)(s^2 + b^2)} \right]$$

7

7. (a) Obtain the Fourier transform of

$$f(x) = x^2, \text{ for } |x| \leq a$$

$$= 0, \text{ for } |x| > a$$

7

(b) Find Fourier Sine transform of $\frac{1}{x}$

7

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3

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BE II (ME)

3

Math.

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME21A (MA) : MATHEMATICS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any **Five** questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) Solve

$$\frac{dx}{dt} - x + 2y = 0; \frac{dy}{dt} - 5x - 3y = 0$$

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1

(Contd.)

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BE II (ME)

3

PMT

SECOND B.E. EXAMINATION - 2024

MECHANICAL ENGINEERING

(III SEMESTER CBCS)

3ME45A (ME) : PRODUCTION MACHINE
TOOLS

Time - Three Hours

Maximum Marks - 70

Note:- (i) Attempt any Five questions.

(ii) Answer should be to the point.

(iii) All questions carry equal marks.

1. (a) What type of work holding devices are generally used in a lathe? 4
- (b) Describe the operation of the quick return motion in a Mechanical Shaper. 5
- (c) Explain briefly the construction of a radial drilling machine with emphasis on how the requisite motions are obtained. 5

J-1438

1

(Contd.)

2. (a) What are the various types of Milling Cutters that are used in milling? 5
- (b) What is the marking system followed in case of grinding wheels? Explain the individual elements of the marking system from the standpoint of the functioning of the wheel. 4
- (c) Explain the concept of location. Also explain the various types of locators used in jigs and fixtures. 5
3. (a) Explain the TIG and MIG systems of arc-welding. Enumerate the applications of each. 7
- (b) Explain the principle underlying the resistance welding process. Enlist the names of products where in the following processes are used:
 - (i) Spot Welding
 - (ii) Seam Welding
 - (iii) Flash Welding
 7
4. (a) What are the various types of automatic lathes in the multi-spindle category? Explain their differences and applications. Is the productivity of multi-spindle machines higher compared to single spindle automatic lathe? 7
- (b) Explain the various operations generally performed in a Sheet-metal shop. 3

J-1438

2

(Contd.)

- (c) Explain the advantages and disadvantages of compound dies over progressive dies. 4
5. (a) Explain Numerical Control (NC) machine tools. What are the classifications of NC machines? 7
- (b) How do you define a Machining Centre? How is cutter compensation given in the case of a machining centre? 7
6. Write short notes on any TWO of the following:
 - (i) Surface grinding
 - (ii) Taper turning methods
 - (iii) Capstan and Turret lathes
 - (iv) Laser Beam Welding
 7X2=14

J-1438

3